

# **Day 1 – Software Testing Fundamentals & Login Testing**

**Name:** Tehzeeb Masood

**Role:** Manual QA Engineer (Learning Portfolio)

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## **Topics Covered**

### **Theory**

- What is Software Testing?
- Why Software Testing is Important
- Bug vs Defect vs Failure
- Functional Testing
- Non-Functional Testing
- Unit Testing
- System Testing
- User Acceptance Testing (UAT)
- Severity vs Priority
- SDLC
- STLC
- SDLC vs STLC

### **SDLC Phases**

1. Requirement Gathering
2. Planning
3. Design
4. Development
5. Testing
6. Deployment
7. Maintenance

### **STLC Phases**

1. Requirement Analysis
2. Test Planning
3. Test Case Development
4. Test Environment Setup
5. Test Execution
6. Test Closure

## **Difference Between SDLC and STLC**

| <b>SDLC</b>                                                    | <b>STLC</b>                           |
|----------------------------------------------------------------|---------------------------------------|
| Software Development Life Cycle                                | Software Testing Life Cycle           |
| Used to develop software                                       | Used to test software                 |
| Followed by Developers, QA, Designers, Business Analysts, etc. | Mainly followed by QA/Testers         |
| Starts from requirements                                       | Starts after QA receives requirements |
| Ends with maintenance                                          | Ends with test closure                |
| Goal: Build software                                           | Goal: Ensure software quality         |

## **Practical Project**

### **Project Name**

Manual Testing of Login Functionality

### **Website**

<https://the-internet.herokuapp.com>

### **Module Tested**

Form Authentication (Login)

### **Test Cases Executed**

| <b>ID</b> | <b>Test Case</b>                    | <b>Expected Result</b>       | <b>Status</b> |
|-----------|-------------------------------------|------------------------------|---------------|
| TC-001    | Valid Username + Valid Password     | Login Successful             | Pass          |
| TC-002    | Invalid Username                    | Error Message                | Pass          |
| TC-003    | Invalid Password                    | Error Message                | Pass          |
| TC-004    | Logout                              | User Logged Out Successfully | Pass          |
| TC-005    | Username with Leading Space         | Login Failed                 | Pass          |
| TC-006    | Username with Trailing Space        | Login Failed                 | Pass          |
| TC-007    | Password with Leading Space         | Login Failed                 | Pass          |
| TC-008    | Password with Trailing Space        | Login Failed                 | Pass          |
| TC-009    | Press Enter Instead of Login Button | Login Successful             | Pass          |

### **Valid Login**

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username  
tomsmith

Password  
.....

Login

Powered by [Elemental Selenium](#)

You logged into a secure area!

## Secure Area

Welcome to the Secure Area. When you are done click logout below.

Logout

Powered by [Elemental Selenium](#)

### Invalid Username

Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

Login

## Invalid Password

ⓘ Your password is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

## Logout

✔ You logged out of the secure area!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

## Username with Leading Space

✔ You logged out of the secure area!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

✖ Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

**Username with Trailing Space**

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

✖ Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

### Password with Leading Space

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

ⓘ Your password is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

### Password with Trailing Space

ⓘ Your password is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

### Press Enter Instead of Login Button

the-internet.herokuapp.com/secure

✔ You logged into a secure area!

## Secure Area

Welcome to the Secure Area. When you are done click logout below.

Logout

## Additional Test Ideas

- Username and password are the same

### Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

 Your password is invalid! 

### Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

- Password with 500 characters

### Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login



ⓘ Your password is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

- Password as "password"

ⓘ Your password is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

- Empty Username

ⓘ Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

- Empty Password

❌ Your password is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

- Both fields empty

❌ Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

- SQL Injection Input

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

ⓘ Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

- HTML Input

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

ⓘ Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

➔ Login

- JavaScript Input

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

🔴 Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

- Emoji Input

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

🔴 Your username is invalid!

## Login Page

This is where you can log into the secure area. Enter *tomsmith* for the username and *SuperSecretPassword!* for the password. If the information is wrong you should see error messages.

Username

Password

 Login

## **Concepts Applied**

- Functional Testing
- Negative Testing
- Input Validation
- Edge Case Testing
- Boundary Value Thinking
- Usability Testing

## **Interview Questions Practiced**

- What is Software Testing?
- What is a Bug?
- What is a Defect?
- What is a Failure?
- Functional vs Non-Functional Testing
- Unit Testing
- System Testing
- UAT
- Severity vs Priority
- SDLC vs STLC
- Can STLC start before SDLC?
- Who writes Test Cases?

## **Learnings**

Today I learned the fundamentals of Software Testing and understood the difference between Bug, Defect, and Failure. I also learned Functional and Non-Functional Testing, SDLC, STLC, Severity, and Priority. Finally, I performed my first real-world manual testing on a live login page by executing multiple positive and negative test scenarios.

## **Skills Gained**

- Software Testing Fundamentals
- Manual Testing
- Login Testing
- Test Scenario Thinking
- Expected vs Actual Result
- QA Mindset
- Basic Edge Case Analysis

## **Day 1 Status**

Completed Successfully

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